M387C_Course_Content

- Lecture 1: Introduction
- Lecture 2: Floating-Point System
- Lecture 3: Condition Number & Abstract Error Analysis for Linear Systems
  - 3.1: Matrix Algebra
    - Basics
      - Matrix-Vector Form
      - Induced Norm
      - Condition Number
    - Properties of K
      - (Prop) 1. K(A)≥1; 2. K(A^-1)=K(A); 3. K(aA)=K(A); 4. K_2(A)=1 if A is orthonormal
      - (Prop) K_2 = \sig_1^\sig_n
      - (Prop) dist_p(A) = K_p(A)^-1
    - (preThm) ||AII<1 => 1/(1+||AII)≤||(I-A)^-1||≤1/(1-||AII)
  - 3.2: Forward a priori analysis
    - Setup
    - (Thm) ||A^-1|| ||dAII <1 => Relative Perturbation Estimate on x
    - (Prop) dA =0 RP Estimate
    - (postThm) ||dAII ≤ \gamma ||AII, ||d bII\≤ \gamma ||bII, if \gamme K(A)<1 => ||x+dxII/IIxII ≤ (1+\gamma K)/(1-\gamma K)
    - (preThm) A,C, R=AC-I, ||RII<1 => ||A^-1|| ≤ ||RII / ||AII ; ||RII/||AII ≤ ||C-A^-1|| ≤ ||CII ||RII / (1-||RII)
  - 3.3: Backward a priori and poteriori
    - Backward a priori analysis
    - A posteriori Analysis
- Direct Method for Linear Systems
  - Gaussian Elimination and LU Decomposition
    - Gaussian Elimination & LU
      - Process
      - Remark
    - Pivoting
      - Row
      - Full
      - Remarks
    - Strassen Matrix-Matrix Multiplication
  - LU and Chelosky Decomposition
    - LU Factorization & Cholesky
      - (Thm) LU exists + unique <=> all principle minors non-singular
        - Theorem
        - Proof
      - (Thm) Cholesky Decomposition
        - Theorem
        - Proof
    - Modern Computer Architectures
      - Memory Hierarchy
      - Data Dependencies
      - Cache Misses
      - Communication Cost
      - Techniques
- Iterative Method for Linear Systems
  - Iterative Method
    - General Idea and Consistency
    - (Thm) Convergence Condition
      - Theorem
      - Proof
    - (preThm) ||B|| ≤ \rho(B) + e
  - Jacobi
    - Jacobi Iteration
    - (Thm) ||B_J||_{\inf} <1 <= A row strictly DD
      - Theorem
      - Proof
    - (Thm) \rho(A)≤||A||
  - Gauss-Seidel
    - Gauss-Seidel Method
    - GS vs Jacobi
    - Example BVP
  - Over-relaxation
    - Relaxation Motivation
    - OR Method
    - SOR Method
    - SSOR
  - Multigrid
    - Motivation
    - Multigrid Method
    - Analysis
      - B
      - Remark
        - Cheap Direct
        - Consistency
        - Residual and Error
        - Types
        - Convergence Notes
          - Smoothing Property
          - Approximation Property
        - Preconditioner
    - Applications
      - Algebraic Multigrid
      - Multilevel Monte Carlo
        - Monte Carlo
        - Multilevel
  - Krylov Subspace Methods
    - Gradient Descent
      - Krylov Subspace
      - GD Derivation
      - GDM
      - Remark
        - Krylov
        - Convergence
    - Conjugate Gradient Descent
      - Motivation
      - Analysis
      - CGDM
      - Remark
      - Error Analysis
        - GD
        - CGD
      - Pre-conditioning
        - Motivation
        - Method P^-1/2 A P^1/2
        - Analysis
        - Choices
          - Jacobi Precondition
          - Multigrid
          - ILU
    - Other Krylov
      - CG only for SPD
      - GMRES
- Eigenvalue Problems
  - Richardson Method
    - General Idea
    - Convergence
  - Power Method
    - Goal
    - Derivation
    - Power Method
    - Convergence Rate
    - (Thm) Validity of Power Method
      - Theorem
      - Proof
    - Error Analysis
      - \lambda_2 << \lambda_1
      - Perturbation Analysis
        - Perturbe
        - (Thm) Bauer-Fike
        - (Thm) Perturbation of y,A,x determines d\lambda
        - Stability of Eigenvalues
        - (Thm) Perturbation of A,x gives \lambda and \ep determines ||x_k(\ep) - x_k||
  - SVD
    - Over/Under-Determined System
    - Over-determined
    - Under-determined
    - SVD
  - Inverse Power Method
    - Goal
    - Inverse Power Method
    - Analysis
  - Eigen Geometry
    - (Thm) A, H, iS => H: \lambda_n≤Re\lambda≤\lambda_1, S: \lambda_n≤Im\lambda≤\lambda_1
    - (Thm) Gershgorin Circles
    - (postThm) First Gershigorin
    - (postThm) Second Gershigorin
  - QR Iteration
    - General QR
      - Simultaneous \lambda_k
      - QR Iteration
      - (Thm) QR convergence
      - Gram-Schmidt
      - (Thm) A real eig and distinct => lim T_k = diag(eigs) upper triangle R
      - O(n^3)
    - Hessenberg QR
      - Householder Transform
        - Transform
        - Reducing to Hessenberg
      - Given Transform
      - QR for Hessenberg to Upper triangle
        - O(n^3) per iter—> O(n^3) then O(n^2) per iter
        - SVD
        - QR with Shift
- Methods for Nonlinear Systems
  - Nonlinear System of Equations
    - Nonlinear Problems
    - Bisection Method
      - IVT
      - Algorithm
      - Error
    - Secant Method
      - Algorithm
      - Properties
    - Fixed-Point Method
      - Fixed-pt Method
      - (Thm) Convergence Condition
    - Newton's Method
      - Newton's Method
      - As Fixed-pt Method
    - Taylor & Error Analysis
      - Derivation from Taylor
      - (Thm) C^2, f'^-1(x^*) exists => Second Order Convergence
      - Properties
    - Quasi-Newton's Method
      - Motivation
      - Generic Quasi-Newton
      - 1D Secant
      - Broyden's Method
  - Fast Methods
    - Direct Methods
      - DFT
      - FFT
    - Approximation Methods
      - FMM
      - SVD Low-rank approx
- Optimization Methods
  - Overveiw
  - Unconstrained
    - Value-Based Methods
      - Golden-section Search Method
      - Nelder-Mead Method
    - Gradient-Based Methods
      - Vanilla GD
      - alpha
      - Line Search
      - Armijo-Goldenstein
      - SGD
    - Hessian-Based Methods
      - Newton
      - Broyden
      - Practical
  - Constrained
    - Penalty Methods
      - Exterior
      - Interior
      - Summary
    - Linear Programming
      - LP
      - Example
      - Dual Problem
      - Numerical Algorithms
        - Simplex
        - IPM
      - 1-\infty Example
    - Nonlinear Problems
      - Compresive Sensing
      - Equality Constraints
        - Lagrange Multiplier
        - Example
      - General Constraints
        - KKT Conditions
        - Example
      - Primal-Dual
        - Equivalence
        - Primal Dual
        - P*≥D*
        - Convex + Slater => Strong Duality
        - KKT as Necessary
        - Complementary Slackness
- Function Approximation
  - Polynomials
    - Polynomial Interpolation
      - Polynomial Interpolation
      - (Thm) m=n => Unique poly
        - Theorem
        - Vandermonde Proof
        - Lagrange Proof
    - Error Estimate
      - (Thm) Error Estimate
        - Theorem
        - Proof
      - Runge's Phenomenon
        - Runge
        - Reason
        - Example
    - Newton Polynomials
      - Newton's Divided Difference
      - Newton Interpolation Polynomial
      - Newton vs Lagrange
      - Example
    - Hermite Polynomials
      - Motivation
      - Lagrange to Hermite
      - Hermite Interpolation
      - Example
    - Interpolation Theory
      - Misc
  - Splines
    - Classic Cubic Splines
      - Cubic Splines
      - Cubic Spline Analysis
      - Cardinal Splines
      - Example
    - B-Splines
      - B-Splines
      - Properties
    - Parametric Splines
      - Motivation
      - Bézier Curve
      - NURBS
    - Applications
      - First Derivative by Linear
      - Second Derivative by Quadratic
      - Error
  - Numerical Integration
    - Numerical Differentiation
      - Motivation
      - Finite Difference
      - Richardson Extrapolation
      - A posteriori error
      - Determining coefs
    - Trapezoidal & Simpson's Method
      - Goal
      - Trapezoidal Rule
        - Method
        - Error
      - Simpson's Method
        - Method
        - Error
    - Gaussian Quadrature
      - Gaussian Quadrature
      - Fixed Nodes
      - Free Nodes
        - Midpt Rule
        - 4 DOF
        - Exactness Condition
        - General
        - Nodes \zeta
    - Newton-Cotes Formula
      - Method
      - Example of Open and Closed
      - Composite Methods
    - Asymptotic Error Expansion
      - Richardson
      - 1-Richardson + Trap = Simpson
      - Romberg
    - Generalized Integral
      - Infinite Domain
        - Goal
        - Method
        - Example
      - Multidimensional
        - Goal
        - Method
        - Example
      - Monte Carlo
        - Motivation
        - Method
        - Properties
        - Generating Random
  - Numerical Differentiation and Integration
    - Orthonormal Families
      - Motivation
        - Interpolation vs Approximation
        - Space & Minimization
        - Galerkin Projection
        - Orthonormal Polys
        - (Thm) Recursive of OP
      - Chebyshev Polynomials
        - Chebyshev
        - Example
        - Orthonormality & PDE
      - Legendre Polynomials
        - Legendre
        - Example
        - Orthonormality & PDE
      - Gaussian Integration
        - Goal
        - Interpolatory Quadrature
        - (Thm) Exactness of Gaussian Integral
      - Fourier Series
        - Fourier Series
        - DFT
        - FFT
        - Pseudo-Spectral Differentiation
    - Wavelets
      - Wavelets
        - Motivation
        - Definition
        - Haar Wavelet
        - Other Wavelets
        - Trade-off
      - Multi-Resolution Analysis
        - Motivation
        - Father functions
        - V_n
        - W_n
        - Decompose
        - Higher Dim
        - Summary
      - Applications
        - MRA
        - Applications
        - Fourier vs Wavelet
        - (Thm) Chebyshev EquiOscillation Thm
    - Applications on Machine Learning